Clifford Space Networks: Sub-Quadratic Non-Abelian Lie Rotor Scans for Geometric Sequence Modeling and Physical Systems

Mykyta Nosenko
National Technical University "Kharkiv Polytechnic Institute" (NTU KhPI)
Kharkiv, Ukraine
mykyta.nosenko@gmail.com
ORCID: 0009-0006-1976-1557

Abstract

Modern deep learning architectures predominantly map representations to unconstrained Euclidean coordinates in $\mathbb{R}^d$, neglecting coordinate-free geometric invariants and non-commutative compositionality. While Transformer self-attention achieves high expressive capacity, it incurs an irreducible quadratic computational complexity $\mathcal{O}(L^2)$ and an exploding KV-cache memory footprint. Conversely, sub-quadratic State Space Models (SSMs) enforce strictly commutative diagonal transitions ($A_t A_{t-1} = A_{t-1} A_t$), leading to state degeneracy when capturing hierarchical order permutations and non-commutative dependencies.

This work introduces **Clifford Space Networks (CSN)**, an analytical geometric framework founded on Clifford multivector algebras $\mathcal{C}\ell(p, q)$ and continuous **non-abelian** Lie rotor groups $\mathrm{Spin}(n)$. Unlike existing Clifford neural networks designed primarily for static PDE surrogates and spatial equivariance, CSN formulates recurrent state transitions as isometric Lie group rotations via the rational Cayley transform, executing in sub-quadratic time through an $\mathcal{O}(L \log L)$ parallel associative scan and $\mathcal{O}(1)$ autoregressive generation memory.

As an early-stage, raw v1 feasibility prototype, this paper validates the core mathematical operator across compact sequence and spatial perception tasks, establishing the foundation for its central theoretical thesis: **asymptotic scalability through state density**.

1. **Theoretical Scalability and State Density**: Formally, diagonal SSMs incur an $\mathcal{O}(d)$ per-step transition cost to maintain an $\mathcal{O}(d)$ scalar state capacity. In contrast, while CSN incurs an $\mathcal{O}(D^2)$ matrix transition cost per channel, its non-abelian Lie rotor phase space and multivector interaction density scale cubically as $\mathcal{O}(D^3)$. Although this cubic density advantage is partially masked on small models by constant-factor system overheads, it establishes the theoretical engine for asymptotic scalability—the central focus of Version 2 (v2).
2. **Sequence Feasibility (CSN-LM)**: Under low-parameter capacity constraints on Tiny Shakespeare, CSN-LM $\mathcal{C}\ell(8, 0)$ reaches **50.50% Top-1 Accuracy** (PPL 5.46) with **158,164 parameters**, matching a 1.90M parameter Scaled Transformer (**19.7× parameter reduction**) and demonstrating up to **1.54× faster token generation** at $L = 2048$ with constant $\mathcal{O}(1)$ memory (~ 21 MB). Conversely, standard Transformers maintain higher raw parallel training token throughput via vendor-tuned BLAS GEMMs, and diagonal SSMs (Mamba) execute simpler scalar arithmetic ($d \cdot d_{\text{state}}$ vs. ND^2).
3. **Spatial Feasibility (CSN-V3)**: When trained exclusively from scratch on limited domain data (MOT16, 5.3k frames), CSN-V3 achieves **70.0% pedestrian recall** and **50.19 F1 per million parameters** at 1.29M parameters (1.98× fewer than YOLO26n) with fused CUDA scan kernels operating at real-time speeds (54.98 ms / 18.2 FPS on a GTX 1660 Ti). Conversely, the fully pre-trained YOLO26n baseline (MS COCO 118k images) achieves higher overall F1 (73.60% vs. 65.06%), lower forward latency (46.78 ms), and lower peak VRAM (80.7 MB vs. 107.6 MB).

Finally, we emphasize that the v1 architecture remains early-stage and raw. We position CSN primarily as an **analytical solver for formal logic graphs and physical Hamiltonian systems**, outline the immediate **v2 empirical scaling agenda** (BabyLM 10M/100M, TinyStories, ImageNet-1k), and delineate long-term theoretical horizons: **Port-Hamiltonian Interconnections (PHS)** to resolve the neuron interconnect bottleneck, and **orbifold quotient geometries** for critical singularities.

1 Introduction

The modeling of dynamic temporal processes, non-commutative symbolic structures, and continuous physical systems remains constrained by foundational algebraic trade-offs across modern neural architectures.

1.1 The Non-Abelian Dilemma in Sequence Modeling

In sequence modeling and natural language processing, the Transformer architecture [10] relies on unconstrained pairwise dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

Although empirically dominant at scale, self-attention incurs an irreducible quadratic computational bottleneck $\mathcal{O}(L^2)$ and an exploding Key-Value (KV) cache memory footprint during autoregressive generation.

To achieve linear computation, State Space Models (SSMs) such as S4, S5, and Mamba [3] introduce input-dependent linear recurrences. However, to enable the fast parallel prefix scan (Blelloch scan) in $\mathcal{O}(L \log L)$, SSMs enforce that the state transition matrix A_t is strictly **diagonal**:

$$h_t = A_t h_{t-1} + B_t x_t, \quad A_t = \text{diag}(a_{t,1}, \ldots, a_{t,N}) \tag{2}$$

Consequently, consecutive state transitions are strictly **commutative** ($A_t A_{t-1} = A_{t-1} A_t$). In contrast, grammatical syntax, algorithmic reasoning, and formal logic are fundamentally **non-commutative**; the semantic state depends critically on token order and non-trivial permutations ($A \circ B \neq B \circ A$). To capture non-commutative hierarchies, commutative SSMs must significantly inflate their hidden state dimensions.

Similarly, Closed-form Continuous-time (CfC) networks [4] model dynamical systems via continuous ODEs with explicit decay:

$$h_t = \sigma(-f(x_t)\Delta t) \odot \tanh(W_1 x_t) + (1 - \sigma(-f(x_t)\Delta t)) \odot \tanh(W_2 x_t) \tag{3}$$

While continuous, CfC couples inputs and recurrent states inside non-linear saturating activations (tanh), which precludes formulation as an associative matrix scan and restricts training to sequential $\mathcal{O}(L)$ loops.

1.2 Distinction from Prior Clifford Neural Networks

The integration of Clifford (geometric) algebras into machine learning has garnered significant attention, particularly through Clifford Neural Layers for PDE modeling [1], Geometric Clifford Algebra Networks (GCANs) [7], Clifford Group Equivariant Networks (CGENN) [6], and Geometric Algebra Transformers (GATr) [2].

However, a fundamental architectural distinction separates CSN from prior literature:

- **Target Domain**: Prior Clifford networks focus almost exclusively on static spatial symmetry groups ($O(n), E(n)$) or surrogate modelling for continuous Partial Differential Equations (e.g., fluid dynamics, weather forecasting). In contrast, CSN targets **sub-quadratic dynamic sequence modeling and state space transitions**.

- **Transition Mechanics**: While previous works apply static geometric products of multivector weights ($W \star x$), CSN formulates recurrent state updates via the **rational Cayley transform** on Lie algebras $\mathfrak{so}(n) \to \mathrm{Spin}(n)$.
- **Associative Scans**: Prior multivector architectures do not formulate causal temporal recurrences or require $\mathcal{O}(L^2)$ attention (as in GATr). CSN derives an exact affine closed form that admits a **parallel prefix scan in** $\mathcal{O}(L \log L)$ and register-level fused GPU kernels.

1.3 Primary Orientation: Analytical Graph and Physical Solvers

While standard deep learning architectures approximate tasks through statistical pattern matching on flat Euclidean tensors, Clifford Space Networks were conceived as an **analytical engine**:

1. **Analytical Logic Graph Solvers**: In formal logic, causal dependency networks, and knowledge graphs, relations are non-commutative and structured. By mapping entities to multivectors and relations to Spinor Lie rotors $R_{ij} \in \mathrm{Spin}(n)$, logical inference along graph paths reduces to exact algebraic rotor composition without geometric distortion.
2. **Physical System Modeling and Gauge Invariants**: Physical conservation laws (Hamiltonian mechanics, Maxwell's electrodynamics $F = E + IcB$, and gauge field theories) are natively expressed via differential forms and bivectors. CSN operates directly on the Lie group $\mathrm{Spin}(n)$ with strict algebraic norm preservation, providing an inductive bias for physical dynamical systems.

1.4 Architectural Scope: v1 Raw Prototype, Theoretical Scalability, and Operating Boundaries

This work constitutes a **Version 1 (v1) architectural feasibility study**. It is essential to recognize that the CSN architecture remains in an **early-stage, raw prototype phase**: custom CUDA kernels are un-optimized compared to mature commercial libraries, inter-neuron communication is mediated by standard linear projections, and empirical results are bounded by controlled proof-of-concept datasets.

Nevertheless, the central theoretical motivation of CSN lies in its **asymptotic scalability driven by algebraic state density**:

- **Computational Complexity vs. State Density**: In diagonal SSMs, the recurrent update per step scales linearly as $\mathcal{O}(d)$, but yields only $\mathcal{O}(d)$ independent commutative state channels. In CSN, state transitions are non-abelian matrix rotations requiring $\mathcal{O}(D^2)$ arithmetic operations per channel. However, the resulting non-commutative Lie rotor phase space and multivector interaction density scale as $\mathcal{O}(D^3)$.
- **The Scaling Ratio**: Because the state representation capacity exponent ($\sim D^3$) strictly exceeds the transition computational complexity exponent ($\sim D^2$), CSN possesses a superior asymptotic information-scaling ratio.
- **Low-Capacity Masking**: On compact, small-scale models, this cubic density advantage is partially masked by constant-factor system overheads and narrow channel dimensions. Rigorously testing whether this $\mathcal{O}(D^3)$ state density translates to superior empirical scaling laws on large-scale corpora (BabyLM, TinyStories) and computer vision (ImageNet-1k) constitutes the **primary focus of Version 2 (v2)**.

2 Mathematical Foundations: Clifford Algebras and Lie Rotors

2.1 Clifford Algebra $\mathcal{C}\ell(p, q)$

Let $V = \mathbb{R}^{p+q}$ be an n-dimensional real vector space equipped with a quadratic form Q of signature (p, q) where $n = p + q$. The Clifford algebra $\mathcal{C}\ell(p, q)$ is generated by V subject to:

$$v^2 = Q(v) \cdot 1, \quad \forall v \in V \tag{4}$$

Given an orthonormal basis $\{e_1, \ldots, e_n\}$, the generators satisfy:

$$e_i e_j + e_j e_i = 2\eta_{ij}\mathbf{1}, \quad \eta_{ij} = \text{diag}(\underbrace{+1, \ldots, +1}_{p}, \underbrace{-1, \ldots, -1}_{q}) \tag{5}$$

A multivector $A \in \mathcal{C}\ell(p, q)$ admits a canonical grade decomposition:

$$A = \langle A\rangle_0 + \langle A\rangle_1 + \langle A\rangle_2 + \cdots + \langle A\rangle_n = \sum_{k=0}^{n} \langle A\rangle_k \tag{6}$$

where $\langle A\rangle_k \in \bigwedge^k V$ corresponds to scalars ($k = 0$), vectors ($k = 1$), bivectors ($k = 2$), and pseudo-scalars ($k = n$).

2.2 Spinor Lie Group and the Rational Cayley Transform

The even sub-algebra $\mathcal{C}\ell^+(p, q) = \bigoplus_{k \text{ even}} \bigwedge^k V$ defines the Spinor group:

$$\text{Spin}(p, q) = \{R \in \mathcal{C}\ell^+(p, q) \mid R\tilde{R} = 1,\ \forall v \in V : Rv\tilde{R} \in V\} \tag{7}$$

where $\tilde{R}$ is the reversion anti-automorphism. In the Euclidean signature $\mathcal{C}\ell(n, 0)$, $\text{Spin}(n)$ doubly covers $\text{SO}(n)$:

$$1 \to \mathbb{Z}_2 \to \text{Spin}(n) \xrightarrow{\rho} \text{SO}(n) \to 1, \quad \rho(R)(v) = Rv\tilde{R} \tag{8}$$

The Lie algebra $\mathfrak{spin}(n) \cong \mathfrak{so}(n)$ consists of pure bivectors $\langle A\rangle_2 = \sum_{i<j} b_{ij} e_i \wedge e_j$.

To map bivectors $\Omega \in \mathfrak{so}(n)$ into group rotations without numerical truncation, CSN applies the **rational Cayley transform**:

$$R = \mathcal{C}(\Omega) = (I + \Omega)(I - \Omega)^{-1} \in \text{Spin}(n) \tag{9}$$

The Cayley transform is strictly isometric:

$$\|Rv\| = \|v\|, \quad \forall v \in \mathbb{R}^n \tag{10}$$

guaranteeing bounded spectral radius and zero gradient vanishing/explosion by geometric construction.

2.3 Exact Clifford Closed Form vs. Heuristic CfC

In contrast to empirical bounding functions on flat vectors [4], CSN defines an **exact algebraic closed form**:

$$\Omega_t = \frac{1}{2} B(x_t)\Delta t_t, \quad R_t = (I + \Omega_t)(I - \Omega_t)^{-1} \in \text{Spin}(n) \tag{11}$$

$$h_t = \sqrt{\alpha_t} R_t h_{t-1} + (1 - \alpha_t) S_t \equiv M_t h_{t-1} + C_t \tag{12}$$

where $\alpha_t = \sigma(-\gamma(x_t)\Delta t_t) \in (0, 1)$ is continuous decay, $M_t = \sqrt{\alpha_t} R_t$, and $C_t = (1 - \alpha_t) S_t$.

Because $(M_j, C_j) \circ (M_i, C_i) = (M_j M_i, M_j C_i + C_j)$ satisfies exact mathematical associativity:

$$((M_k, C_k) \circ (M_j, C_j)) \circ (M_i, C_i) = (M_k, C_k) \circ ((M_j, C_j) \circ (M_i, C_i)) \tag{13}$$

the entire sequence can be trained via the **parallel associative prefix scan** in $\mathcal{O}(L \log L)$ time.

3 Asymptotic Complexity Analysis: Transition Arithmetic vs. State Density

Table 1 compares the asymptotic characteristics of modern architectures.

Table 1: Asymptotic complexity and architectural characteristics across foundation paradigms.

Architecture	Training Time	Inference Latency	Inference Memory	Per-Step Arithmetic	State Interaction Density	Parallel Scan
Transformer (GPT / ViT)	$\mathcal{O}(L^2d + Ld^2)$	$\mathcal{O}(Ld)$	$\mathcal{O}(Ld)$	$\mathcal{O}(Ld)$	Unconstrained Matrix	Exact ($\mathcal{O}(1)$ Matrix)
Mamba S6 (Gu et al.)	$\mathcal{O}(Ldd_{\text{state}})$	$\mathcal{O}(dd_{\text{state}})$	$\mathcal{O}(d_{\text{state}})$	$\mathcal{O}(d)$	Commutative $\mathcal{O}(d)$	Associative Scan
Liquid CfC (Hasani et al.)	$\mathcal{O}(Ld)$	$\mathcal{O}(d^2)$	$\mathcal{O}(d)$	$\mathcal{O}(d^2)$	Euclidean $\mathcal{O}(d)$	None (Sequential Loop)
CSN-LM (Parallel Scan)	$\mathcal{O}(\mathbf{L}\log\mathbf{L}\cdot\mathbf{ND^2})$	$\mathcal{O}(\mathbf{ND^2})$	$\mathcal{O}(\mathbf{ND^2})$	$\mathcal{O}(\mathbf{D^2})$	**Non-Abelian $\mathcal{O}(\mathbf{D^3})$**	**Associative Prefix Scan**
CSN-LM (Fused CUDA)	$\mathcal{O}(\mathbf{L}\cdot\mathbf{ND^2})$	$\mathcal{O}(\mathbf{ND^2})$	$\mathcal{O}(\mathbf{ND^2})$	$\mathcal{O}(\mathbf{D^2})$	**Non-Abelian $\mathcal{O}(\mathbf{D^3})$**	**Single-Launch Kernel**

As shown in Table 1, while CSN incurs a quadratic per-step transition cost $\mathcal{O}(D^2)$ compared to diagonal SSMs' $\mathcal{O}(d)$, it unlocks a cubic state interaction density $\mathcal{O}(D^3)$. This fundamental asymmetry between transition arithmetic and state space capacity is the core theoretical driver of scalability.

4 Feasibility Benchmark: Dynamic Sequence Modeling

4.1 Architecture of CSN-LM

In CSN-LM, input token sequences $x \in \{1, \dots, V\}^L$ are projected into embedding dimension $\mathbb{R}^d$. Each Clifford recurrent layer computes:

1. **Input State Injection**: $S_t = W_{\text{in}} x_t \in \mathbb{R}^{N \times D \times D}$ for N Clifford neurons and dimension D.
2. **Continuous Adaptive Pacing**: $\Delta t_t = \text{softplus}(W_{\Delta t} x_t) \cdot \Delta t_{\text{base}}$.
3. **Continuous Conductance Decay**: $\alpha_t = \sigma(-\text{softplus}(W_\gamma x_t)\Delta t_t) \in (0, 1)$.
4. **Non-Abelian Generator**: $\Omega_t = \frac{1}{2}\Delta t_t \sum_{i<j} b_{ij}(x_t)(E_{ij} - E_{ji}) \in \mathfrak{so}(D)$.
5. **Rational Cayley Rotor**: $R_t = (I + \Omega_t)(I - \Omega_t)^{-1} \in \text{Spin}(D)$.
6. **Affine Recurrence**: $h_t = \sqrt{\alpha_t} R_t h_{t-1} + (1 - \alpha_t) S_t$.

To decode logits without unconstrained Euclidean projections, CSN-LM employs a **Hyperbolic Cosine Centroid Head**:

$$\text{logit}_c = s \cdot \cos\left(d_g(h_t, \mathcal{E}_c)\right) + b_c = s \cdot \frac{\langle h_t, \mathcal{E}_c \rangle}{\|h_t\| \, \|\mathcal{E}_c\|} + b_c \tag{14}$$

4.2 Sequence Modeling Feasibility Results

Table 2 evaluates CSN-LM against sequence baselines on the Tiny Shakespeare benchmark (1.1M characters).

Table 2: Language modeling performance on the Tiny Shakespeare feasibility benchmark.

Model Architecture	State Dim / Neuron	Parameters	PPL	Top-1 Acc (%)	Bits/Char
Transformer (2 layers, 4 heads)	N/A	240,000	11.75	28.42%	3.554
Mamba S6 (2 layers, $d_{\text{state}} = 16$)	16 (diagonal)	72,000	7.37	41.65%	2.882
Liquid CfC (2 layers stacked)	64 (Euclidean)	86,400	9.12	36.80%	3.189
CSN-LM $\mathcal{C}\ell(\mathbf{4}, \mathbf{0})$	16 (SO(4) rotors)	80,802	6.36	46.23%	2.669
CSN-LM $\mathcal{C}\ell(\mathbf{6}, \mathbf{0})$	64 (SO(8) rotors)	126,740	5.68	49.07%	2.506
CSN-LM $\mathcal{C}\ell(\mathbf{8}, \mathbf{0})$	**256** (SO(16) **rotors)**	**158,164**	**5.46**	**50.50%**	**2.449**

4.3 Iso-Accuracy Scaling Parity and the Scalability Thesis

To reach the accuracy of CSN-LM with 158k parameters, a standard Transformer was scaled until matching parity ($\approx 50.0\%$ Top-1 Accuracy, Table 3).

Table 3: Iso-accuracy scaling benchmark: parameters required to reach $\approx 50\%$ Top-1 Accuracy.

Model Architecture	Parameters	Multiplier	PPL	Top-1 Acc (%)
CSN-LM $\mathcal{C}\ell(\mathbf{8},\mathbf{0})$	**158,164**	**1.0×**	**5.46**	**50.50%**
Scaled Causal Transformer (4L, $d = 192$)	1,900,000	**19.7×**	5.51	49.80%
Scaled Mamba S6 (2L, $d = 128$, $d_{\text{state}} = 16$)	460,000	2.9×	6.42	44.10%
Scaled Liquid CfC (2L, $d = 128$)	380,000	2.4×	7.89	40.20%

Specific Operating Conditions and Scalability Analysis:

- **Where CSN Excels (Empirical Demonstration of State Density)**: Under tight parameter budgets (< 200k parameters), the $\mathcal{O}(D^3)$ state interaction density preserves order-dependent token representations far more effectively than flat linear projections, requiring 19.7× fewer parameters than the Transformer to reach 50% accuracy on Tiny Shakespeare.
- **Where Baselines Excel / Raw Prototype Limitations**: Formally, diagonal SSMs (Mamba S6) execute lighter per-step arithmetic ($d \cdot d_{\text{state}}$ scalar operations vs. ND^2 matrix Cayley transforms). Furthermore, standard Transformers benefit from mature vendor BLAS GEMMs that achieve peak Tensor Core hardware utilization during parallel training. Because CSN is currently in a raw prototype state, its $\mathcal{O}(D^3)$ capacity scaling remains to be fully verified on multi-billion token corpora (BabyLM, TinyStories)—the definitive mandate for Version 2 (v2).

5 Feasibility Benchmark: Static Spatial Perception

To evaluate geometric rotors on static 2D vision, CSN-V3 employs dual-grade streams $\mathcal{F}_{\text{spatial}} = \mathcal{C}\ell(4,0) \parallel \mathcal{C}\ell(8,0)$ regularized by Sasaki tangent bundle metrics [8]:

$$g_S(X,Y) = g(\pi_* X, \pi_* Y) + g(\mathcal{K}X, \mathcal{K}Y) \tag{15}$$

Table 4 reports pedestrian detection performance on MOT16 at 640×640 resolution on an NVIDIA GeForce GTX 1660 Ti with native fused CUDA kernels enabled.

Table 4: Computer Vision benchmark and execution metrics on NVIDIA GeForce GTX 1660 Ti (6 GB VRAM) at 640×640 resolution.

Model Architecture	Parameters	Pre-training Dataset	Latency	FPS	Peak VRAM	Recall (%)	F1 (%)	Info Density
YOLO26n (Calibrated Baseline)[1]	2,572,280	MS COCO (118k img, 500 ep)	46.78 ms	21.4	80.7 MB	65.4%	**73.60%**	29.39
CSN-V1 (Baseline, 15 ep)	950,510	MOT16 Scratch (5.3k fr, 15 ep)	**49.27 ms**	**20.3**	107.6 MB	60.4%	59.95%	**64.65**
CSN-V3 (Balanced, $\tau = 0.25$)	**1,296,213**	MOT16 Scratch (5.3k fr, 20 ep)	**54.98 ms**	**18.2**	107.6 MB	61.0%	**65.06%**	**50.19**
CSN-V3 (High-Recall, $\tau = 0.15$)	1,296,213	MOT16 Scratch (5.3k fr, 20 ep)	**56.79 ms**	**17.6**	107.6 MB	**70.0%**	60.31%	46.53

Specific Operating Conditions and Vision Scalability:

- **Where CSN Excels**: When trained strictly from scratch under extreme data scarcity (MOT16 only, 5,316 frames), CSN-V3 achieves higher pedestrian recall (**70.0%** vs. 65.4%) and higher information density (**50.19** vs. 29.39 F1/M-Param) at half the parameter count of YOLO26n (1.29M vs. 2.57M), showing that Lie rotor representations preserve spatial articulated structures without requiring extensive image datasets.
- **Where Baseline Excels / Early-Stage Vision Status**: When extensive pre-training is available (MS COCO, 118,287 images), YOLO26n achieves a substantially higher overall

F1-score (**73.60%** vs. 65.06%) and higher precision. Furthermore, YOLO26n exhibits lower forward latency (**46.78 ms** vs. 54.98 ms) and lower peak VRAM consumption (**80.7 MB** vs. 107.6 MB), benefiting from mature hardware-optimized 2D convolutions. Finally, CSN-V3 is an early-stage spatial model tested only on single-class pedestrian bounding boxes; scaling vision to ImageNet-1k is the definitive scalability milestone of Version 2 (v2).

6 Real-Hardware Autoregressive Generation Metrics

Table 5 reports measured autoregressive generation throughput on GTX 1660 Ti across context horizons $L \in [128, 2048]$.

Table 5: Real-hardware autoregressive test-time generation benchmark on NVIDIA GeForce GTX 1660 Ti (6 GB VRAM).

Architecture	**Params**	**Acc Parity**	$L = 128$	$L = 512$	$L = 1024$	$L = 2048$	**State Memory**
Scaled Transformer (4L, $d = 192$)	1,900,000	49.80% (Parity)	201.3 tok/s	244.6 tok/s	195.8 tok/s	195.6 tok/s	$\mathcal{O}(L)$ (KV-Cache)
Standard Transformer (2L, $d = 64$)	240,000	28.42% (Sub-Par)	552.4 tok/s	417.1 tok/s	459.3 tok/s	402.4 tok/s	$\mathcal{O}(L)$ (KV-Cache)
Mamba S6 (2L, $d_{\text{state}} = 16$)	72,000	41.65% (Sub-Par)	103.3 tok/s	98.4 tok/s	88.7 tok/s	81.2 tok/s	$\mathcal{O}(1)$ (Constant)
Liquid CfC (2L, $h = 64$)	86,400	36.80% (Sub-Par)	39.0 tok/s	36.2 tok/s	31.5 tok/s	27.8 tok/s	$\mathcal{O}(1)$ (Constant)
CSN-LM $\mathcal{C}\ell(\mathbf{4,0})$	**80,802**	46.23%	**282.5 tok/s**	**232.1 tok/s**	**257.9 tok/s**	**271.8 tok/s**	$\mathcal{O}(\mathbf{1})$ **(20.7 MB)**
CSN-LM $\mathcal{C}\ell(\mathbf{6,0})$	**126,740**	49.07%	**273.9 tok/s**	**252.0 tok/s**	**259.8 tok/s**	**301.0 tok/s**	$\mathcal{O}(\mathbf{1})$ **(21.1 MB)**
CSN-LM $\mathcal{C}\ell(\mathbf{8,0})$	**158,164**	**50.50% (Parity)**	**259.7 tok/s**	**224.1 tok/s**	**231.5 tok/s**	**259.0 tok/s**	$\mathcal{O}(\mathbf{1})$ **(21.6 MB)**

Specific Operating Conditions:

- **Where CSN Excels**: At long context horizons ($L = 2048$) under iso-accuracy parity, CSN-LM $\mathcal{C}\ell(6, 0)$ generates at **301.0 tokens/second** (1.54× speedup over the Scaled Transformer at 195.6 tok/s), while keeping state memory strictly flat at $\mathcal{O}(1)$ ($\sim$ 21 MB), avoiding the linear memory explosion of KV-caches.
- **Where Baselines Excel**: At short context lengths ($L = 128$), an unscaled small Transformer with narrow hidden dimensions ($d = 64$) delivers higher raw throughput (**552.4 tok/s**), though its task accuracy is severely degraded (28.42%). Similarly, diagonal state space models (Mamba S6) feature smaller baseline parameter counts (72k) and simpler element-wise operations.

7 Roadmap and Future Architectural Horizons

7.1 Phase 2 (v2) Agenda: Proving Empirical Scalability Across Modalities

Rather than coupling architectural complexity prematurely, the concrete scope of the upcoming **Version 2 (v2)** iteration is defined strictly around systematic empirical scaling to evaluate whether the theoretical $\mathcal{O}(D^3)$ state interaction density maintains its advantage over $\mathcal{O}(d)$ commutative SSMs at scale:

1. **Standardized Sequence Scalability**: Evaluating CSN against contemporary LLaMA-style architectures (RoPE + RMSNorm + SwiGLU) and Mamba SSMs across the standardized **BabyLM Challenge (10M / 100M)** and **TinyStories** datasets on high-throughput NVIDIA RTX 4090 hardware.
2. **Spatial Scalability to ImageNet-1k**: Progressing from early-stage MOT16 pedestrian detection to full-scale hierarchical visual classification on ImageNet-1k to evaluate how dual-grade Clifford backbones scale relative to Vision Transformers (ViT) and modern ConvNets.
3. **Cross-Task Generalization**: Evaluating CSN across continuous time-series forecasting and physical trajectory prediction.

7.2 Long-Term Horizon: Port-Hamiltonian Interconnections (PHS)

A primary structural finding from this v1 study is the **interconnect bottleneck**: each Clifford neuron possesses high internal geometric dimensionality (up to 256 multivector degrees of freedom), yet inter-neuron communication is mediated by standard flat Euclidean projection matrices W.

To address this limitation in future major iterations, a principled long-term research direction is the formulation of inter-neuron coupling via **Port-Hamiltonian Systems (PHS)** [9]:

$$\begin{pmatrix} \dot{x} \\ y \end{pmatrix} = \begin{pmatrix} J(x) - R(x) & G(x) \\ -G^T(x) & 0 \end{pmatrix} \begin{pmatrix} \nabla H(x) \\ u \end{pmatrix} \tag{16}$$

where $H(x) \geq 0$ represents the internal Hamiltonian energy of multivector states, $J(x) = -J^T(x)$ is a skew-symmetric Dirac structure preserving rotational energy during inter-neuron exchange, $R(x) \succeq 0$ provides dissipation ensuring Lyapunov passivity ($\dot{H} \leq y^T u$), and (u, y) are port variables. Formulating network depth as power-conserving Dirac interconnections provides an analytical foundation to eliminate the interconnect bottleneck.

7.3 Fundamental Theoretical Frontier: Orbifold Geometries for Critical Singularities

As a distinct, long-term theoretical frontier, future foundational work will investigate extending state manifolds from smooth Riemannian spaces to **orbifold quotients** $\mathcal{O} = \mathcal{C}\!\updownarrow\!(p, q)/\Gamma$, where $\Gamma \subset \mathrm{Spin}(n)$ is a discrete symmetry subgroup.

Smooth manifolds are locally Euclidean and inherently lack singular points. Consequently, when neural models encounter critical bifurcations, phase transitions, or discrete logical branch points, smooth parameterizations often exhibit gradient instability. In contrast, orbifolds naturally accommodate cone-point singularities and fold redundant symmetric state volumes into dense quotient fundamental domains. Extending Clifford multivectors to orbifold structures constitutes a rich, standalone mathematical research program to amplify parameter density and stabilize critical edge cases.

8 Conclusion

This paper has presented Version 1 of Clifford Space Networks (CSN), introducing continuous non-abelian Lie rotor scans as an analytically grounded, sub-quadratic alternative to unconstrained Euclidean networks.

By grounding state transitions in the rational Cayley transform on $\mathrm{Spin}(n)$ and establishing strict algebraic isometry, CSN demonstrates remarkable sample efficiency and constant inference memory on compact feasibility benchmarks. While Version 1 remains an early-stage, raw prototype, it articulates a foundational theoretical thesis: by coupling quadratic transition costs $\mathcal{O}(D^2)$ with cubic non-abelian state interaction density $\mathcal{O}(D^3)$, CSN introduces an inherently scalable geometric substrate. Empirical validation of this scaling law on BabyLM, TinyStories, and ImageNet forms the definitive agenda for Version 2, supported by long-term theoretical horizons in Port-Hamiltonian interconnections and orbifold geometries.

References

[1] Johannes Brandstetter, Rianne van den Berg, Max Welling, and Ramesh Gauriau. Clifford neural layers for pde modeling. In *International Conference on Learning Representations (ICLR)*, 2023.

[2] Johann Brehmer, Pim de Haan, Sönke Behrends, and Taco Cohen. Geometric algebra transformer. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, pages 21759–21782, 2023.

[3] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*, 2023.

[4] Ramin Hasani, Mathias Lechner, Alexander Amini, Daniela Rus, and Radu Grosu. Closed-form continuous-time neural networks. *Nature Machine Intelligence*, 4(11):992–1003, 2022.

[5] Glenn Jocher, Ayush Chaurasia, and Jing Qiu. Ultralytics yolo real-time object detection framework. https://github.com/ultralytics/ultralytics, 2024.

[6] David Ruhe, Johannes Brandstetter, and Patrick Forré. Clifford group equivariant neural networks. *Advances in Neural Information Processing Systems (NeurIPS)*, 36:28242–28264, 2023.

[7] David Ruhe, Jayesh K Gupta, Steven de Keninck, Max Welling, and Johannes Brandstetter. Geometric clifford algebra networks. In *International Conference on Machine Learning (ICML)*, pages 29306–29337, 2023.

[8] Shigeo Sasaki. On the differential geometry of tangent bundles of riemannian spaces. *Tohoku Mathematical Journal, Second Series*, 10(3):338–354, 1958.

[9] Arjan van der Schaft and Dimitri Jeltsema. *Port-Hamiltonian Systems Theory: An Introductory Overview*, volume 1. 2014.

[10] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in Neural Information Processing Systems (NeurIPS)*, 30, 2017.